Vinod Kumar Dhanavath

DevOps / Cloud Engineer

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Seasoned DevOps and Cloud Engineer with **7+** years of experience in designing, implementing, and managing scalable cloud infrastructures and CI/CD pipelines. Seeking to leverage deep expertise in AWS/Azure/GCP, containerization, automation, and infrastructure-as-code to optimize deployment processes, enhance system reliability, and drive operational efficiency in a forward-thinking organization.

PROFILE SUMMARY

- Expert in cloud platforms including **AWS, Azure, and GCP**, with hands-on experience in compute, storage, networking, and cloud-native services like **ECS, Fargate, Lambda, Azure Functions, and Logic Apps**.
- Skilled in **CI/CD pipeline design and management** using **Jenkins, GitLab CI/CD, Azure DevOps**, enabling faster and reliable software delivery.
- Extensive experience with **Infrastructure as Code (IaC)** tools including **Terraform, CloudFormation, and Pulumi** for automated, repeatable infrastructure provisioning.
- Proficient in **containerization and orchestration** using **Docker, Kubernetes, EKS, and AKS**, with expertise in microservices and serverless architectures.
- Strong knowledge of **configuration management and automation** with **Ansible, Chef, and Puppet**, plus scripting expertise in **Python, Bash, and PowerShell** for operational efficiency.
- Demonstrated expertise in **cloud security and compliance** (IAM, VPC, security groups, HIPAA, GDPR, SOC 2) and implementing monitoring solutions using **Prometheus, Grafana, ELK, CloudWatch, and Azure Monitor**.
- Experienced in **high-availability deployment strategies** including blue-green, canary, and rolling updates to ensure zero-downtime releases.
- Adept at **cloud cost optimization, performance tuning, and resource management**, reducing operational expenses while improving efficiency.
- Strong leadership and collaboration skills, working with cross-functional teams to implement DevOps best practices, enforce governance policies, and drive enterprise-scale cloud adoption.

PROFESSIONAL EXPERIENCE

Client: National Bank, Ottawa, Canada

Role: Azure DevOps Engineer **Jan 2025 - Current**

<u>Key Responsibilities and Achievements:</u>

- Led cloud migration of enterprise applications from on-premises to **Azure**, designing scalable, highly available, and cost-optimized solutions while enforcing **cloud governance and compliance**.
- Developed and maintained **Infrastructure as Code (IaC)** using **Terraform, ARM templates, and PowerShell scripts** for automated provisioning of VMs, networking, storage, and security resources.
- Built and managed **Azure Kubernetes Service (AKS)** clusters for containerized microservices applications, implementing high availability, self-healing, and optimized container orchestration.
- Designed and implemented **CI/CD pipelines** using **Azure DevOps**, automating deployment to AKS and Azure resources with YAML-based scripts, enabling zero-downtime releases.
- Automated administrative and operational tasks through **Azure Automation Runbooks, PowerShell, and Graphical Workflows**, reducing manual intervention and improving operational efficiency.
- Configured and managed **Azure AD Connect, ADFS authentication flows, and secure identity solutions**, ensuring enterprise-grade access management and compliance.
- Implemented **network security best practices**, including NSGs, Private Endpoints, load balancers, firewall rules, disk encryption, and Key Encryption Keys (KEK) for secure Azure deployments.
- Monitored and optimized cloud infrastructure using **Azure Monitor, Log Analytics, and Application Insights**, establishing proactive alerting, logging, and performance optimization strategies.

- Deployed and managed **databases and storage solutions**, including MySQL, SQL Server, Cosmos DB, Azure Storage Accounts, VHDs, and Availability Sets for reliable, high-performance applications.
- Secured applications and optimized performance with **Nginx reverse proxy, SSL/TLS encryption, Ingress API Gateway, load balancing strategies**, and integrated logging/monitoring for enterprise-grade production environments.

Technologies Used: Azure Cloud (ARM), Terraform, Azure DevOps, PowerShell, Azure Kubernetes Services (AKS), Azure Active Directory (AD), Azure SQL, Azure Data Factory, Azure Storage, Azure Virtual Networks, Azure Recovery Vault, YAML, Nginx, SSL/TLS.

Client: Intact Insurance, Ottawa, Canada

Role: AWS DevOps Engineer **Jan 2023 – Dec 2024**

Key Responsibilities and Achievements:

- Architected, deployed, and managed **AWS cloud infrastructure** including EC2, VPC, S3, RDS, IAM, CloudTrail, and CloudFormation, ensuring **high availability, security, and regulatory compliance** for healthcare applications. Implemented **MFA, IAM policies, and role-based access controls** to safeguard sensitive patient data.
- Designed and automated **CI/CD pipelines** using **Docker, AWS CodePipeline, CodeBuild, CodeDeploy, and Jenkins**, reducing deployment time, minimizing errors, and enabling consistent application releases.
- Implemented **scalable and resilient infrastructure** using Elastic Load Balancers (ELB), Auto Scaling groups, and CloudWatch monitoring, ensuring optimal performance and rapid incident detection.
- Developed **serverless solutions** with AWS Lambda and API Gateway to automate event-driven processes, data ingestion, and backend workflows, improving operational efficiency and reducing manual tasks.
- Optimized **AWS resource utilization and cost management** by leveraging automation with Ansible, scheduling resource scaling, and enforcing governance policies.
- Migrated on-premises workloads to AWS using **Server Migration Service (SMS), Database Migration Service (DMS), and Snowball**, implementing lift-and-shift strategies with minimal downtime and ensuring data integrity.
- Managed **database and storage solutions** including Amazon RDS, Redshift, and S3, ensuring automated backups, snapshots, lifecycle policies, performance tuning, and data security, supporting analytics and reporting requirements.

Technologies Used: AWS Cloud, EC2, VPC, RDS, S3, IAM, CloudFormation, CloudTrail, Docker, GitHub, AWS CodePipeline, CodeBuild, CodeDeploy, ELB, Auto Scaling, CloudWatch, Redshift, Python, Ansible, AWS CLI, AWS Lambda, API Gateway.

Client: (Genpact) Mahindra Finance, Mumbai, India

Role: DevOps Engineer **Aug 2019 - Nov 2022**

Key Responsibilities and Achievements:

- Architected and implemented **end-to-end data ingestion and integration pipelines** using **Azure Data Factory, Azure SQL Database, and Data Warehouse**, optimizing data flow, transformation, and storage for analytics and reporting.
- Designed, automated, and maintained **CI/CD pipelines** with **Jenkins, Docker, and Python scripts**, managing Git repositories, automating package deployment, and ensuring reliable application delivery to Tomcat servers.
- Managed and optimized **cloud infrastructure on Azure and Kubernetes platforms**, leveraging **VMs, Ansible automation, OpenShift, Cloud Foundry, and Helm charts** to ensure high availability, scalability, and fault-tolerant microservices deployments.
- Implemented and maintained **containerization and orchestration solutions** using **Docker, Docker Swarm, and Kubernetes**, including image management, load balancing, and automated deployment pipelines for enterprise-grade applications.
- Customized **project management and DevOps workflows** by configuring **Jira workflows, permission schemes, and integrating Docker Slave Nodes into Jenkins**, improving collaboration, deployment efficiency, and operational governance.

Technologies Used: Azure Data Factory, Azure SQL Database, Data Warehouse, Python, Git, Jenkins, Tomcat, Azure Virtual Machines, Ansible, Terraform, Cobbler, OpenShift, Kubernetes, Cloud Foundry, Docker, Helm, JIRA, Docker Hub, Docker

Client: Jio, Mumbai, India

Role: Cloud Engineer **Jan 2017 - May 2019**

Key Responsibilities and Achievements:

- Designed, deployed, and managed **AWS cloud services** including Lambda, EC2, S3, RDS, IAM, and Route 53, implementing **automated workflows, cross-region routing, and robust security controls** to ensure scalable and secure cloud environments.
- Built and maintained **CI/CD pipelines** using **Jenkins, Gradle, and Bitbucket**, automating testing, packaging, and deployment of microservices to **Docker registries and Kubernetes clusters** for high availability and rapid delivery.
- Implemented **containerization and orchestration solutions** using **Docker, ECS, and Kubernetes**, managing Helm charts, manifest files, and container replicas to ensure efficient deployment, lifecycle management, and operational consistency.
- Integrated **monitoring and observability tools** like Prometheus and Grafana to track application and infrastructure performance, enabling proactive issue detection and optimized resource utilization.
- Automated deployment, scaling, and configuration management for cloud-based applications, streamlining multi-environment releases and ensuring compliance with **enterprise cloud security and governance standards**.

Technologies Used: AWS CodePipeline, CloudFormation, AWS Lambda, Jenkins, AWS S3, Route 53, Security Groups, Network ACLs, IAM, EC2, RDS, Bitbucket, Docker, Kubernetes, Gradle, Artifact Repository, Prometheus, Grafana, Helm, AWS ECS

KEY SKILLS

- **Cloud Platforms:** AWS, Microsoft Azure, GCP
- **Containerization & Orchestration:** Docker, Kubernetes (EKS, AKS, GKE), Helm Charts, ECS, Fargate
- **Infrastructure as Code (IaC):** Terraform, CloudFormation, Pulumi, ARM Templates, Ansible Playbooks
- **CI/CD Tools:** Jenkins, GitLab CI/CD, Azure DevOps Pipelines, GitHub Actions, Bamboo, ArgoCD
- **Version Control:** Git (GitHub, GitLab, Bitbucket), SVN
- **Configuration Management:** Ansible, Puppet, Chef
- **Scripting & Automation:** Python, Bash, PowerShell, Groovy
- **Monitoring & Logging:** Prometheus, Grafana, ELK Stack (Elasticsearch, Logstash, Kibana), AWS CloudWatch, Splunk, Datadog, New Relic
- **Networking & Security:** VPC, Subnets, Security Groups, Load Balancers, DNS, CDN (CloudFront, Akamai), Firewalls, VPN, IAM, KMS, Secrets Manager
- **Databases:** MySQL, PostgreSQL, SQL Server, DynamoDB, Cosmos DB, MongoDB, Redshift, BigQuery
- **Build & Artifact Management:** Maven, Gradle, Nexus Repository, Artifactory
- **DevSecOps / Security Tools:** SonarQube, Checkmarx, Snyk, Aqua Security, HashiCorp Vault
- **Workflow Orchestration / ETL:** Apache Airflow, AWS Step Functions, Azure Data Factory
- **Messaging & Web Servers:** Apache Kafka, RabbitMQ, Nginx, Apache HTTP Server
- **Project Management & Collaboration:** Jira, Confluence, Trello, Slack, Microsoft Teams

EDUCATION

Masters: Osmania University, Hyderabad (Aug 2014 – Sep 2016)